Gravity above the surface

antocciphysics | TEACHER KEY | CPA

Learning target: Use distance from a planet’s center to compare fields at altitude.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 4; 3 course days later.*
3 N; divide by 25 because force follows inverse square separation.

02. Explain a worked example (worked)
Radius R plus altitude 2 R makes 3 R; using h alone gives the wrong denominator.

03. Look and explain (model)
h=R; center separation r=2 R.

04. Build the solution (practice)
3 N/kg.

05. Use the idea (reason)
Astronaut and spacecraft fall together; little support force. Floating is not evidence of absent gravity.

06. Choose a method (mixed) *Review: Unit 1, Day 17; 12 course days later.*
At rest: friction 24 N opposite the push. Sliding: net force 24 N with the push.

07. Exit check (exit)
1 N/kg.

08. STRETCH: Apply it another way (stretch)
Center distance doubles to 2 R, so altitude is R.

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01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 4; 3 course days later.*
Fixed masses: 3 r. Both masses enlarged: 15 r gives the required force fraction.

02. Explain a worked example (worked)
Radius R plus altitude 2 R makes 3 R; using h alone gives the wrong denominator.

03. Represent and interpret (model)
h=R; center separation r=2 R.

04. Solve and justify (practice)
8.68 N/kg; 2.45 N/kg.

05. Analyze the reasoning (reason)
Astronaut and spacecraft fall together; little support force. Floating is not evidence of absent gravity.

06. Choose a method (mixed) *Review: Unit 1, Day 17; 12 course days later.*
N=39.2 N; static limit 19.6 N. Low push: friction 8 N, a=0. High push: kinetic friction 15.68 N, a=2.08 m/s².

07. Exit check (exit)
h=R; h=2 R.

08. STRETCH: Apply it another way (stretch)
Center distance r/R=√10; altitude h/R=√10-1≈2.16.